DO NOW — RECALL

C# Login Project · Code a Hash Table

Name: _____ Class: _____ Date: _____

1. A login check uses validUser && validPassword . What is the result when validUser is true and validPassword is false? Explain.

2. A D-type flip-flop currently stores Q = 1. D changes to 0 but there is no triggering clock edge. What is Q?

3. A queue receives A, B and C in that order. Remove one item, then enqueue D. List the remaining items from front to rear.

4. A five-bucket hash table uses h(k) = k MOD 5 with linear probing. Insert keys 7, 12 and 17 in that order into an empty table. State each final index.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

False. AND (&&) is true only when both conditions are true.

2. Model answer

Q remains 1. The input is sampled only when the triggering clock edge occurs.

3. Model answer

B, C, D. A leaves first because a queue is FIFO.

4. Model answer

7 hashes to 2 and is stored at 2. 12 also hashes to 2, so probes to 3. 17 also hashes to 2; 2 and 3 are occupied, so it goes to 4.

By the end, you can…

• Write a simple hash function and store usernames in an array.

• Resolve collisions using linear probing and search for usernames.

• Write synoptic MCQs on linked structures and hash tables.

Vocabulary — know these terms

hash table — A table using a hash function to locate data from a key.

hash function — A repeatable mapping from key to bucket index.

bucket — A location in the hash table.

collision — Two keys map to the same bucket.

linear probing — Resolve collisions by checking successive buckets.

key — The value used to look up an entry.

lookup — Find a stored item using its key.

load factor — Fraction of table buckets occupied.

NEW LEARNING

Read, annotate and explain

Implement hashing and collisions

Classroom demonstration only. Never type, collect or store real passwords.

The problem: more than one user

A single username comparison is not scalable. We need a way to store and find many usernames.

What is a hash function?

Convert the username to a number, then use % tableSize to choose a bucket index. Our small function is deliberately simple for learning.

What if two names choose one bucket?

That is a collision. Use linear probing: try the next bucket, wrapping to the beginning when necessary.

Lookups must use the same rule

Recalculate the starting bucket from the username, then probe through occupied buckets until the username or an empty bucket is found.

Hash-table lookup versus password hashing

The hash-table function locates an account. It does NOT securely protect a password. Use invented demo passwords only.

MCQ revision

Write four precise MCQs on linked lists, stacks, queues and trees; extend with BST and hash-table MCQs if time allows.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Implement hashing and collisions

Classroom demonstration only. Never type, collect or store real passwords.



"ab" and "ba" both start at index 6

A username passes through a hash function and maps to a numbered bucket. Collisions require probing.

I DO — WORKED EXAMPLE

Follow the worked method

Build a real hash-table index

Create the buckets

```
const int SIZE = 7;
string?[] usernames = new string?[SIZE];
```

Write the index function

```
static int Hash(string key, int size)
{
    int total = 0;
    foreach (char letter in key)
        total += (int)letter;
    return total % size;
}
```

Explain the limitation

```
// Hash("ab", 7) == Hash("ba", 7)
// Same total -> collision.
// This is an educational hash function,
// NOT a secure password hash.
```

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Insert with linear probing

What happens when a bucket is already occupied?

Method / working:

2. Search using the same probing sequence

How can we find a name after a collision?

Method / working:

3. Predict a collision

With table size 7, why do 'ab' and 'ba' choose the same initial index?

Method / working:

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Insert with linear probing What happens when a bucket is already occupied? Explain your solution first

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 2: Search using the same probing sequence How can we find a name after a collision?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 3: Predict a collision With table size 7, why do 'ab' and 'ba' choose the same initial index?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

CODING 1 · INSERT
Create a seven-bucket string array, implement Hash(), and insert at least four test usernames. Print each chosen bucket index.

Question 2

CODING 2 · COLLISION
Register 'ab' and 'ba'. Show that they collide and use linear probing to store both in different buckets.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

CODING 3 · LOOKUP
Write a search that follows the same probing sequence. Return found/not found, including for a username never inserted.

Question 4

SYNOPTIC MCQ BANK
Write MCQs on linked lists, directed/undirected graphs, stacks, queues, trees, binary search trees and hash tables. Make a six-question draft; each needs four options and a written explanation.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A hash-table function based on adding character codes is suitable for protecting real passwords.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Insert with linear probing

CODE:

```
int index = Hash(username, SIZE);

while (usernames[index] != null)

{

index = (index + 1) % SIZE;

}

usernames[index] = username;
```

IMPROVE: Explain one correction to your own reasoning.

Guided 2 — original CHECK

Explain your solution first

How can we find a name after a collision?

Search using the same probing sequence

CODE:

```
int index = Hash(username, SIZE);

int checkedBuckets = 0;

while (checkedBuckets < SIZE && usernames[index] != null)

{

if (usernames[index] == username) break;

index = (index + 1) % SIZE;

checkedBuckets++;

}
```

IMPROVE: Explain one correction to your own reasoning.

Guided 3 — original CHECK

Explain your solution first

With table size 7, why do 'ab' and 'ba' choose the same initial index?

Predict a collision

Both strings contain the same character codes, so the total is identical. Linear probing moves the second username to the next free bucket.

IMPROVE: Explain one correction to your own reasoning.